



Complete Schedule of XXV ISSSD

September 22-26, 2025

Local Venue: CDTN - Belo Horizonte, Brazil

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XXV International Symposium on Solid State Dosimetry

22-26 SEPTEMBER 2025 - BELO HORIZONTE, BRAZIL



General Schedule

General Schedule					Hour			
Pre-Symposium - Monday, 22 Sep					8h			
Registration	Development of PET Radiopharmaceuticals - Dr. Marina Bicalho Silveira Dr. Isabel Dregely Dr. Emerson Bernardes Soares (Spectrum)		Introduction to Radiation Detectors - Dr. Marco A Polo Labarrios (Lattice)		9h			
					9h30			
					10h			
					10h30			
					11h			
COFFE BREAK					11h30			
					12h			
LUNCH					13h			
					13h30			
					14h			
COFFE BREAK	Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico - Dr. Silvia S Hidalgo Tobón (Spectrum)		Solid State Dosimetry: Principles and Applications & OSL Advanced Materials - Dr. Ivonne Berenice Lozano Rojas Dr. Alida Tamar-Gabor (Lattice)		14h30			
					15h			
					15h30			
					16h			
					16h30			
					17h			
Tuesday, 23 Sep		Wednesday, 24 Sep		Thursday, 25 Sep	Friday, 26 Sep	Hour		
Registration		Oral Sessions (Spectrum and Lattice)	Poster Session: 10h00 to 12h30 - Referees for evaluation (Hall, Virtual Room)	Invited Speaker: Luciana Carvalheira (Lumina)	Invited Speaker: Rolf Behrens (Lumina)	8h		
Opening Ceremony (Lumina)				Invited Speaker: Bernardo Maranhão Dantas Maria Jose Neves (Lumina)	Invited Speaker: Susana Lalic (Lumina)	8h40 9h		
COFFE BREAK				COFFE BREAK		Invited Speaker: Thiago Viana Miranda Lima (Lumina)	9h20	
Invited Speaker: Wilson Calvo (Lumina)		Oral Sessions (Spectrum and Lattice)	Poster Session: 10h00 to 12h30 - Referees for evaluation (Hall, Virtual Room)	COFFE BREAK		10h		
Invited Speaker: Juan Azorín Nieto (Lumina)				Invited Speaker: Ademir Lugo (Lumina)	Oral Sessions (Lumina and Spectrum)	10h30		
				Invited Speaker: Richard Hugtenburg (Lumina)	Invited Speaker: Tarcisio Campos and Alberto Campos (Lumina)	11h 11h30		
LUNCH		LUNCH		LUNCH		12h30		
Oral Sessions (Lumina and Spectrum)	Poster Session: 14h30 to 16h30 - Referees for evaluation (Hall)	Roundtable: Radon in Environment: Tracer Applications and Public Health Risks (Lattice)	Technical Visit	Oral Sessions - (Spectrum and Lattice)	Oral Sessions (Lumina, Spectrum and Lattice)	13h30		
	Roundtable: AI's challenges and Opportunities: Connecting research, industry and society (Lumina)				Invited Speaker: Helio Yoriyaz (Lumina)	Poster Session: 14h30 to 17h - Referees for evaluation - (Hall)	Oral Sessions (Lumina, Spectrum and Lattice)	14h
COFFE BREAK					Invited Speaker: Georgia Joana (Lumina)		Invited Speaker: Yvone Maria Mascarenhas (Lumina)	14h30
				COFFE BREAK		15h		
	Poster Session: 14h30 to 16h30 - Referees for evaluation (Hall)	Invited Speaker: Silvia Tobón (Lumina)		COFFE BREAK		15h30		
				Helen Khoury Mentorship WIN-Brasil (Spectrum)	Invited Speaker: Carmen Bueno (Lumina)	16h		
Welcome & Honors Night (Lumina)				Oral Sessions (Lumina and Lattice)	Closing Session: Honor award & Helen Khoury award for best works (Lumina)	16h30		
						17h		
						17h30		
						18h00		
						20h00		

Monday, 09/22/2025

08:00 *Registration*

Courses Pre-Symposium		
	Spectrum Auditorium	Lattice Auditorium
	Chair: Marina Bicalho	Chair: Álvaro M. L. Gómez
10:30 COFFE BREAK	08:00 Dr. Marina Bicalho, Dr. Isabel Dregely and Dr. Emerson Bernardes <i>Development of PET Radiopharmaceuticals</i>	08:00 Dr. Marco Antonio Polo La-barrios <i>Introdution to Radiation Detectors</i>

12:00 LUNCH

Courses Pre-Symposium		
	Spectrum Auditorium	Lattice Auditorium
	Chair: Angel Ramirez	Chair: Carlos E. Velasques
15:00 COFFE BREAK	13:00 Dr. Silvia S Hidalgo Tobón <i>Importancia de la física en una imagen por resonancia magnética en el diagnóstico clínico</i>	13:00 Dr. Ivonne Berenice Lozano Rojas and Dr. Alida Tamar-Gabor <i>Solid State Dosimetry: Principles and Applications & OSL Advanced Materials</i>

Tuesday, 09/23/2025

08:00 *Registration*

Opening Cerimony - Lumina Auditorium

09:00 Opening Cerimony - **Bruno Melo Mendes ((ISSSD 2025 - Chair))** and **Amenônia Maria Ferreira Pinto** - Centro do Desenvolvimento da Tecnologia Nuclear (CDTN/CNEN), **Henrique Resende Martins** - Vice-director of Escola de Engenharia of Universidade Federal de Minas Gerais (UFMG), **Wilson Aparecido Parejo Calvo** - Comissão Nacional de Energia Nuclear (CNEN), **Silvia S. Hidalgo Tobón ((ISSSD 2025 - Chair))** and **Juan Azorin Nieto ((ISSSD 2025 - Chair))** - Universidad Autónoma Metropolitana (UAM), **Telma C. F. Fonseca (ISSSD 2025 - Chair)** - Universidade Federal de Minas Gerais (UFMG).

10:00 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Silvia Hidalgo

10:30 *Invited Speaker* – Wilson Aparecido Parejo Calvo,
Current Status of Strategic Projects at CNEN.

11:30 *Invited Speaker* – Juan Azorin Nieto,
Thermoluminescence and Its Application in Radiation Dosimetry.

12:30 LUNCH

Roundtable Session (Lattice Auditorium)

13:30 Radon in Environment: Tracer Applications and Public Health Risks
Moderator: Talita Santos,
Panelists: Elydio Soares, Laura Takahashi, and Nathan da Silva

Radiobiology

Spectrum Auditorium
Chair: Tarcísio Campos

Luminescent Materials/ Applications of Thermoluminescence
Lumina Auditorium
Chair: Susana Lalic

13:30 Jhon Brandon Felix Ferreira
G07-002 - *Low-Dose Neonatal Ionizing Radiation Induces Modest, Sex-Specific Changes in Adult Mouse Behavior and Hippocampal Oxidative Stress*

13:45 Lucas Siqueira de Lima Neto
G07-013 - *Radiometric Analysis by gamma spectrometry of copaiba oil using a Hyper*

14:00 Iris Bomfim Martins Ferreira
G07-015 - *Radiometric analysis of the bivalve mollusk Perna*

13:30 João Pedro
VanellusRad - *The First Brazilian Real-Time IoT Electronic Dosimeter for Radiation Monitoring*

14:00 Ángel Ramírez Luna
G01-002 - *Afinando la Cronología Relativa de Cerámicas de la Zona de Playa Vicente, Veracruz, México Aplicando el Método de Datación Absoluta por Termoluminiscencia*

Roundtable Session (Lumina Auditorium)

Chair: Telma C. F. Fonseca

14:30 AI's Challenges and Opportunities: Connecting Research, Industry, and Society
Moderator: Telma C. F. Fonseca,
Panelists: Rita Sebastião, Melina Castro and Ana Carolina

13:30 to 16:30 - Poster Session (Main Hall)

**Note: 14:30 to 16:30 - Referees for evaluation
In-Person**

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

- G01-001** - Olania Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLL-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*
- G01-011** - Samira Batista da Silva, *Enzymatic biocatalysis of alginate spheres and iodinated amino acids for applications in embolization guided by SPECT*
- G01-026** - Roseli Künzel, *A low-cost integrated system for thermoluminescence and optically stimulated luminescence measurements*
- G01-029** - Yuri Avelino Costa, *Analysis of the average geometric parameters of the closed pores of the fine aggregate matrix using X-ray micro-computed tomography and linear regression*
- G01-032** - Danilo O. Junot, *Crab shell biowaste composites as luminescent radiation detectors*
- G01-035** - Luiz Matheus de Souza Aguiar Leite, *Development of a radiochromic film using a polyacetylene compound*
- G01-036** - Thalles Oliveira Campagnani, *Development of a test bench using a CsI/SiPM detector for mass flow measurement in industrial processes*
- G01-038** - Ketrin Regina Souza, *Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership*
- G01-040** - Raissa Alexia Camargo Guassu, *Estimation of absorbed dose in critical organs during interventional procedures: experimental and computational approach*
- G01-041** - Danilo O. Junot, *Influence of precursor purity on the synthesis and TL/OSL response of CaSO₄-based radiation detectors*
- G01-044** - Vinicius Costa Silva, *Radiological characterization of the waters feeding the Camorim Reservoir by alpha-beta – Pedra Branca State Park*
- G01-045** - Margarete Cristina Guimarães, Thessa Cristina Alonso, *Standardization of gamma irradiation procedures for the preservation of bibliographic heritage*

G5 - Radiological Protection

- G05-002** - Wallacy Viana, *Analysis of the characteristics of the Ingeo™ 3D850 polymer in the context of its use in visual inspections in the TRIGA IPR-R1 nuclear reactor*
- G05-003** - Adriana de Souza Medeiros Batista, *Assessment of volcanic rock adsorbents for the remediation of acid mine drainage in uranium mining*
- G05-006** - Daniel de Castro Pacheco, *Development of a distributed monitoring system for active radiological protection*
- G05-007** - Cássio Miri Oliveira, *Dose evaluation and suggestion of typical values for head CT headache protocols: a preliminary study*
- G05-015** - Thamyra Cybelle Vieira dos Santos, *Quality assurance in ⁹⁰Y radioembolization: an intercomparison protocol for activity optimization*
- G05-017** - Fábio Lima Guimarães, *Radiometric survey at the Stone House Lookout, Saquarema-RJ*
- G05-019** - Marcela Moraes Freitas, *Structural and compositional changes in zeolitic materials exposed to uranium mine effluents: implications for environmental radioprotection*
- G05-022** - Bianca Rossini Marques, *Validation of analytical methods using gamma spectrometry applied to the radiometric characterization of monazite sand*

G8 - Luminescent materials

- G08-004** - Divanizia do Nascimento Souza, *Development of high-sensitivity CaSO₄:Tm,Mn detectors for thermoluminescent and optically stimulated luminescent dosimetry*
- G08-007** - Roseli Künzel, *Effect of silver doping concentrations on the structural and luminescence properties of CaMoO₄ powders*
- G08-008** - Paula Thays Santos Malachias, *Investigation of the luminescent properties of zirconia-enriched dental resins*
- G08-009** - Bruno Araújo, *Optically stimulated luminescence (OSL) characteristics of seven different color*

types of quartz samples under 405 nm stimulation

G08-010 - Rafael dos Santos Viana, *Promotion of the Compton effect of ^{137}Cs photons to enhance the efficiency of radioluminescent batteries*

G08-012 - Patrícia de Lara Antonio, *PTTL and PTOSL of TLD-100 exposed to ^{60}Co and lighting with LEDs*

G08-013 - Daniele Gonçalves Mesquita, *Radiation Detection of Lead-Free BaZrO_3 Perovskite Scintillators*

G08-014 - Claudete Roberta Evangelista, *Feasibility of Retrospective Dosimetry Using Ceramic Material from Spark Plugs*

G08-015 - Anderson Manoel Bezerra da Silva, *Structural, Morphological, Luminescent, and Dosimetric Properties of $\text{CaSO}_4:\text{Yb,Mn}$ Phosphors*

G08-016 - André Felipe de Oliveira, *Tailored theranostic nanocomposites: mesoporous silica, gold nanoparticles, and gadolinium complexes for cancer diagnosis and treatment*

G08-020 - Jéssica Pauline Nunes Marinho, *Potential theranostic nanoparticles of HAp-Gd^{3+} with curcumin and folic acid for cancer therapy*

15:30 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Ivonne B. L. Rojas

16:00 *Invited Speaker* – Silvia S. Hidalgo Tobón,

State of the art of magnetic resonance imaging: from fundamental physics to clinical applications.

17:00 - 20:00 Welcome & Honors Night - Lumina Auditorium

- For the Ceremony: Bruno M. Mendes and Telma C. F. Fonseca
- Award Ceremony for Honored Researchers: Dr. Tarcísio P. R. de Campos and Dr. Teógenes Augusto da Silva
- Cultural Performance by Aruanda Group
- Welcome Cocktail Gathering

Wednesday, 09/24/2025

Applications of Thermoluminescence
Spectrum Auditorium
Chair: Andrea Mantuano

08:00 Rafael Cogollo

G01-004 - *Dosimetric characteristics and kinetic analysis of pure and cerium-doped alumina (α - Al_2O_3) for low-dose thermoluminescent dosimetry applications*

08:15 Héctor Maya

G01-006 - *Kinetic analysis method for general order TL glow curves*

08:30 Natalia Barbosa Gonzaga Mendes

G01-024 - *Using gamma spectrometry as a tool for assessing residual dose in irradiated topaz*

08:45 Alberto Mizrahy Campos

G01-027 - *Random walk of neutrons in a rotational environment*

09:00 Enrique Alejandro Chávez Camacho

G01-030 - *Analysis of the effects of gamma radiation on amaranth seeds and sprouts: a design of experiments (DoE) approach*

09:15 Paula Selvatice Pereira Teles

G01-031 - *Application of the MCNP code in dose verification for stereotactic radiosurgery (SRS)*

09:30 Alexander C. F. da Silva

G01-028 - *Absorbed Dose Evaluation in Breast Tissues Using Heterogeneous Models*

10:00 COFFEE BREAK

Radiation Sources
Spectrum Auditorium
Chair: Jhonny Benavente

10:30 Roberto Méndez Villafañe

G06-001 - *Development of a detailed model with MCNP of the Neutron Standards Laboratory at CIEMAT and validation with a Bonner Sphere Spectrometer System*

10:45 Nayelly Nikole Díaz Gutiérrez

G06-002 - *Estimación del término fuente de un irradiador gamma de ^{60}Co utilizando el código MCNP*

11:00 Roberto Méndez Villafañe

G06-003 - *Neutron measurements around the*

Dosimetry
Lattice Auditorium
Chair: Ana Carolina Costa

08:00 Berenice Guadalupe Olmos Grimaldo

G02-006 - *Medición de radón en la zona del Peñón Blanco, Salinas de Hidalgo, San Luis Potosí, México*

08:15 Renato Monaco Nunes

G02-018 - *A Monte Carlo-based software for photon transport in water using the CUDA platform*

08:30 Iasmin Vieira Nishibayaski

G02-019 - *A new formulation of Fricke gel dosimeter development using salicylic acid*

08:45 Camilla da Silva Sampaio

G02-026 - *Assessing the applicability of TDCR counting for nonpure beta emitters: the case of ^{210}Pb in liquid scintillation cocktails*

09:00 Laura Cardoso Takahashi

G02-040 - *Development of an integrated national database for effective dose calculation from environmental radiation exposure in Brazil*

09:15 Laisa de Almeida Neves

G05-023 - *Analysis of Natural Radioactivity in Beach Sands Along the Rio de Janeiro State Coast by Gamma Spectrometry*

Medical Physics
Lattice Auditorium
Chair: Lucas Paixão

10:30 Sara Suely Ventura da Silva Lima

G04-004 - *Comparative Monitoring of Lens, Thyroid, and Whole Body in Interventional Cardiology Procedures*

10:45 Prycylla Gomes Creazolla

G02-089 - *Volumetric dose evaluation using 6 MV linear accelerators for blood irradiation*

11:00 Juraci Passos dos Reis Junior

G04-011 - *An AI revolution in radiotherapy in SUS: The example of Mário Kroeff Hospital*

research reactor RA-6 at Centro Atómico Bariloche with a Bonner Sphere Spectrometer

11:15 Ismael Teixeira Vicente

G06-004 - *Alpha particle spectrometry and Monte Carlo evaluation of ^{241}Am sources for use in Elastic Recoil Detection Analysis (ERDA)*

11:30 Estela Maria de Oliveira

G06-006 - *Comparative study of gamma efficiency curves in HPGe detector with different calibration standards*

11:45 Paulo Alberto Lima da Cruz

G06-010 - *Evaluation of thermal neutron-induced nuclear reactions in niobium foil by HPGe spectrometry and FLUKA/FLAIR Monte Carlo simulation*

12:00 Roos S. de Freitas Dam

G06-012 - *Neutron Source Applied for Eccentric Scale Prediction in Oil Pipelines using Deep Learning*

11:15 Luis Alberto Perez-Martinez

G04-013 - *Automated detection and characterization of renal calculi using U-Net and Hounsfield unit analysis in computed tomography scans*

11:30 Teodoro Córdova Fraga

G04-019 - *COVID largo y fibrosis pulmonar: clasificación basada en tomografía en población mexicana*

11:45 Natasha Briggs e Silva

G04-020 - *Determination of the mass attenuation coefficient of thermoset and thermoplastic resins at energies used in conventional mammography*

08:00 to 12:30 - Poster Session (Main Hall)

Note: 10:00 to 12:30 - Referees for evaluation In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-003 - Emilia Suárez, *Development and Assessment of Perovskite-based X-Ray Detectors for In Vivo Dosimetry Applications*

G02-012 - Aline Fabiane Gonçalves de Oliveira, *Geospatial assessment of radionuclide dispersion and associated effective dose modeled with ARTM View: a case study at CDTN's radiopharmaceutical unit*

G02-017 - Tales Noujaim Silveira, *^{68}Ga Production with the GE PETtrace-8 Cyclotron: a Monte Carlo study*

G02-020 - Leanderson Pereira Cordeiro, *AI-based segmentation performance and implications for organ-level dosimetry in radionuclide therapy*

G02-021 - José Marques Lopes, *An Environmental Radiometric Assessment Based on Airborne Gamma Spectrometry Data in the State of Rio de Janeiro*

G02-023 - Hugo Leonardo Lemos Silva, *Analytical and simulation methods in determining radiation exposure time of healthcare waste*

G02-027 - Alessandra Vaz dos Anjos Santos, *Automated System for Visual Detection of Personal Dosimeters Using Convolutional Neural Networks*

G02-028 - Tiago Macedo da Cunha e Silva, *Bayesian-Optimized Monte Carlo Modeling of CyberKnife Collimators and Source Parameters for Small-Field Dosimetry*

G02-030 - Heloísa Milena Caldeira Rhodes, *Comparison of Methods for Determining the Half-Value Layer (HVL) in Computed Tomography Equipment*

G02-031 - Nicolle Marques Cardoso Ignacio, *Comparison of the Dosimetric Properties of EBT-3 and EBT-4 Radiochromic Films for Application in Hemocomponent Dosimetry*

G02-032 - Fernanda Rocha Cavalcante, *Comparison of X-ray spectral correction software for a CdTe detector*

G02-033 - Guilherme Gazolla Santana, *Computational Modeling of an Anthropomorphic Phantom for Applications in Aerospace Dosimetry*

G02-034 - José Marques Lopes, *Computational Simulation of a NaI(Tl) Detector and Definition of Its Main Parameters for Application in a Neutron Activation Analysis Laboratory*

- G02-035** - Lucas Beloti Silva, *Computational Tools for Automatic Conversion of Microscopy Images into Voxelized Cellular Phantoms*
- G02-043** - João Vinícius Batista Valença, *Dose estimation in organs of a pediatric patient undergoing brain radiotherapy via Monte Carlo simulation*
- G02-044** - Wallacy Viana, *Dose Rate Estimation by Extracting and Analyzing Colors from an Image Generated by a CMOS Image Sensor*
- G02-045** - Guilherme Cavalcante de Albuquerque Souza, *Dosimetric comparison of methods for obtaining dose profiles in head CT scans using 120 kV X-ray beams*
- G02-046** - Frederico Possato Tagliaferro, *Dosimetry Of Electret Microphones Irradiated By X-Rays*
- G02-047** - Josemary A. C. Gonçalves, *Energy Dependence of an Epitaxial Diode for Radioprotection Dosimetry*
- G02-048** - Enzo Cortez, *Estimating the dose rate constant for Iodine-131 via Monte Carlo simulation: a first step towards pediatric radioiodine therapy shielding evaluation*
- G02-050** - João Vinícius Batista Valença, *Evaluating Radiation Dose in Veterinary Professionals: MCNPX-Based Simulation of Equine Radiography Exposure*
- G02-051** - Camila Nunes Carvalho Souza, *Evaluating the Performance of Bayesian Optimization in High-Dose-Rate Brachytherapy Planning*
- G02-052** - Maria Rita Pascoal de Souza Barros, *Evaluation of Absorbed Dose Uniformity in Irradiated Foods with Different Packaging Geometries Using PMMA Dosimetry*
- G02-054** - Gabriel Gomes do Nascimento, *Evaluation of the reproducibility of a new lens dosimeter holder*
- G02-061** - João Victor Ramos Oliveira, *Influence of detector geometry on small-field radiotherapy simulations*
- G02-066** - Daniel de Castro Pacheco, *Modeling and validation of a Whole Body Counter for in vivo measurements using CAD and MCNP6*
- G02-070** - Greiciane de Jesus Cesário, *Monte Carlo Simulation of Depth Dose in Mammography Using Geant4 and 3D-Printed ABS and PETG Phantoms*
- G02-071** - Kyssylla Monnyelle Araujo Silva, *Multielectrode Array Devices for dosimetry and dose enhancement factor analysis in nanoparticle radiotherapy*
- G02-072** - Mateus de Souza Resende, *^{99m}Tc Radiolabeled Papain Nanoparticle: in silico Dosimetry Evaluation using DM BRA phantom and the experimental biodistribution data*
- G02-074** - Henrique Chaves Gulino, *New Low Cost Process of PIN Diode Fabrication for Radiation Detection*
- G02-077** - Carmen C. Bueno, *Online response of a homemade diode dosimetry system in a mobile 500-700 keV electron beam facility*
- G02-078** - Anderson Manoel Bezerra da Silva, *Optimization of Mn²⁺ and Tb³⁺ Concentrations in CaSO₄:Mn,Tb and Advanced Investigation of its TL/OSL Dosimetric Properties*
- G02-079** - Rômulo Monteiro Lopes, *Parametrization and Validation of a HPGe Detector by Monte Carlo Simulation*
- G02-080** - Hugo Leonardo Lemos Silva, *Assessing Photon and Electron Ranges in Healthcare Waste for Radiation Sterilization*

Note: 10:00 to 12:30 - Referees for evaluation
On-line

G1 - Applications of thermoluminescence (dosimetry, dating, industrial, etc.)

- G01-007** - Lenin Estuardo Cevallos Robalino, *Monte Carlo characterization of neutron personal dosimeter based on TLD-600 and TLD-700 irradiated with ²⁵²Cf neutron source on the ISO-phantom at LPN-CIEMAT*
- G01-010** - Abel Fernando Martínez Cruz, *Thermoluminescence of MgO Chemically Modified with Nd and Li Ions, Obtained by Glycine-based Solution Combustion Synthesis*
- G01-013** - Hassan Salah, *Assessing Patient Exposure in PET/CT scans: A Head and Neck Dose Parameter Study*
- G01-017** - Diego Eduardo Calvo-Medina, *Characterization of Beryllium Oxide as TLD of Clinical Use*
- G01-019** - Mohammed, *Evaluation of Radiation Dose Reduction Strategies in Whole-Body PET/CT Imaging*
- G01-021** - Hassan Salah, *Optimizing Radiation Dose Parameters in PSMA PET/CT for Prostate Cancer: Balancing Diagnostic Efficacy and Patient Safety*

- G01-022** - Abdelmoneim Sulieman, *Patient-Specific Dose Optimization in Breast PET/CT: Towards ALARA-Compliant Protocols*
- G01-033** - Edwar Alonzo Canaza Mamani, *Datación OSL-SAR y caracterización tecnológica de las cerámicas arqueológicas de los grupos de tradición ceramista Tupiguarani: Pirajuí I y Arataka II, Pernambuco, Brasil*
- G01-034** - David Alexander Urbina Leal, *Design and Analysis of FDM Prototypes for Gold-198 Nanoparticles*

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

- G02-001** - Alejandro Ferreira Tapia, *Caracterización Física y Computacional de Dispositivos Modeladores de Radiación Electromagnética mediante Simulación Monte Carlo*
- G02-002** - Ernesto Saucedo-Hernandez, *Determination by GEANT 4 the biological damage caused by hadrons and carbon ions using a borated water phantom*
- G02-004** - José Edgardo Arellano Hernández, *Estudio de la respuesta termoluminiscente de $\text{Na}_2\text{O}-\text{Y}_2\text{O}_3\text{-P}_2\text{O}_5\text{-SiO}_2$ irradiados con Electrones*
- G02-008** - Woody Alem Wanderley Araripe Farias, *A platform for creating computational exposure models: personalized dosimetry with Holmium-166*
- G02-009** - Woody Alem Wanderley Araripe Farias, *All about MARTIN: a BREP phantom with blood and lymphatic vessels applied to Yttrium-90 radioembolization*
- G02-013** - Tiago da Silva Jornada, *Longitudinal Monitoring of Occupational Radiation Exposure in Healthcare Professionals: Profile of Critical Hospital Sectors and Guidance for Radiological Protection Measures*
- G02-015** - Jardel Lemos Thalhofer, *Radiometric Profile Analysis of Soils from the São José de Itaboraí Paleontological Natural Park*
- G02-016** - Marcelo Franklin da Silva Santos, *Synthesis and Luminescent Characterization of $\text{CaSO}_4\text{:Tm,Li}$ via Slow Evaporation, Co-precipitation, and Solid-State Diffusion Routes*
- G02-022** - Akemi Yagui, *Organ Dose Estimation in Pediatric Abdomen and Pelvis Radiographs Using Caldose X Monte Carlo Simulations*
- G02-024** - Caio César Santos Valeriano, *Application of Molecular Docking for Structural Modeling of Molecules in simulations with TOPAS-nBio: A Benchmark Against Conventional Material Definitions*
- G02-025** - Ranulfo da Silva Dias, *Assembly and Geant4-based computational simulation of a Helium-3 detector for ground-level cosmic radiation monitoring*
- G02-029** - Daniel Vieira Lamas, *Characterization of the Ubatuba Formation: Integration of Gamma Radiometry, Total Organic Carbon, and Well Logs from the Upper Cretaceous to Lower Paleogene Intervals of the Campos Basin*
- G02-037** - Paulo Sérgio de Abreu Junior, *Cytogenetic assessment of ionizing radiation exposure in *Allium cepa* using micronucleus frequency: a comparative study between Antarctic and control conditions*
- G02-038** - Ana Laura Burin, *Development of a Computational Phantom of the Vertebral Column for Dosimetric Investigations*
- G02-039** - Leonardo da Silva Boia, *Development of a Dedicated System for Generating Input Files for the MCNP and Adaptation of a Phantom with Edema for Simulation of Prostate Brachytherapy with ^{125}I*
- G02-059** - Lucas Fabricio de Araujo, *In Silico Investigation of the Impact of Gold Nanoparticles on the Radiosensitivity of 3D HeLa Cells Using the Monte Carlo Method*
- G02-060** - Gabriela Rodé de Assis da Silva, *In Situ Gamma-Ray Spectrometric Mapping of Beach Sand Radioactivity in Niterói, Brazil*
- G02-067** - Pedro Henrique Avelino de Andrade, *Monte Carlo Modeling of the BPW34 Photodiode for X-ray Dosimetry*
- G02-073** - Nadia Rodrigues dos Santos, *Natural Radionuclide Activity and Radiological Risk Assessment in Brazilian Coffee Samples*
- G02-086** - Marcelo Luís dos Santos Filho, *Study of $\text{CaSO}_4\text{:Tm,Li}$ Based Luminescent Composites Applied to TL/OSL Dosimetry in Mammography*
- G02-087** - Leonardo da Silva Boia, *MCNP simulation of the effects of prostatic edema in LDR brachytherapy with ^{125}I using the adapted ICRP 110 voxel male phantom*
- G02-093** - Daniel V. Lamas, *Quantification of Gamma Radiation on Drilling Cuttings Through HPGe Detector: Comparative Study With Well Logs Spectrometry Data*

G3 - Ionizing and non-ionizing radiation

- G03-005** - Samara Mendes Matos, *Effect of Neutron Activation in the Biodegradable Chitosan polymer as-*

sociated with Gold Nanoparticles.

G03-010 - Angel Garcia-Duran, *Simulador de detector de Germanio de alta pureza con fuentes radiactivas, embebido en FPGA*

G4 - Medical Physics

G04-001 - Nicolás Eugenio Martín, *Adaptation and performance of CdTe-based XFCT detection system using Ag-tacers for Biomedical Applications*

G04-007 - Priscila Santos Amorim, *Implementation of free software for managing imaging protocols*

G04-008 - Angelica Viridiana Roman Martinez, *Magnetic Resonance Imaging Assessment of Corpus Callosum Volume in Pediatric Epilepsy Patients*

G04-009 - Jaime Torres Juárez, *Optimization of the Pulse Sequence for Visualization of Myocardial Fiber Architecture Using Diffusion Tensor Imaging in the Study of Cardiomyopathies*

G04-012 - Gisell Ruiz Boiset, *Assessment of radiologically tissue-equivalent materials for breast imaging phantoms using X-ray spectrometry*

G04-015 - Anna Luiza Fraga da Silveira, *Characterization of the new FX-PEO dosimetric formulation in radiotherapy beams*

G04-024 - Jesús Emmanuel Mares Ponce, *Evaluation of elemental Iron content in pharmaceutical formulations using magnetic susceptibility technique*

G04-026 - Camila Hitomi Murata, *Evaluation of the affect on performance of a 3T MRI by the proximity of a subway station and an adjacent MRI scanner*

G04-027 - Carolina Salinas Domján, *Experimental characterization of magnetic fields for dosimetric modeling with Monte Carlo methods in MR-Linac radiotherapy*

G04-029 - Ana Carolina Costa da Silva, *Global Usability of Mobile CT Scanners in Rapid Stroke Diagnosis: A Literature Review*

G5 - Radiological Protection

G05-001 - Natalia Matos Orozco, *Diseño y cálculo de blindajes para garantizar la protección radiológica en una radiofarmacia productora de radiofármacos*

G05-008 - Akemi Yagui, *Evaluation of radiation exposure in pediatric interventional cardiology procedures*

G05-012 - Sitti Yani, *Monte Carlo simulation of flux and secondary particle analysis of gadolinium, cadmium, boron, and lithium-based neutron radiation shielding materials*

G05-013 - Tiago Amaral de Lima, *Physicochemical, Radiometric, and Elemental Characterization of Commercial Integral Grape Juices by Gamma Spectrometry and X-ray Fluorescence*

G05-018 - Watila Lins Silva, *Reference Measurements of Thermal Neutron Flux Using a Cadmium-Encapsulated ^6LiI (Eu) Detector at the Neutron Metrology Laboratory*

G05-020 - Fábio Sabará Dias, *Survey of radiation exposure from portable X-ray fluorescence*

G6 - Radiation Sources

G06-005 - Gláucia de Oliveira Barreto, *Axial neutron flux mapping in the Argonauta Research Reactor at IEN-CNEN using neutron activation and γ -ray spectrometry techniques*

G06-007 - Stanislav Svetachev, *Determination of detection limits of elements in sediment and rock standards for INAA using the IGR reactor*

G06-008 - João Pedro Carvalho Andrade, *Determination of the mass attenuation coefficient of polymers using Monte Carlo N-Particle simulation*

G06-011 - Gabriel Tura Echeverria, *Identification and quantification of K, Na and Mn in textured soy protein sold in the city of Rio de Janeiro using the Argonauta research reactor of IEN/CNEN and the techniques of instrumental neutron activation analysis and γ -ray spectrometry*

G7 - Radiobiology

G07-007 - Emilio Romero Muñoz, *Germination response of 'Micro-Tom' to controlled magnetization by using a Rodin coil*

G07-010 - Raúl Ken Delgado Velázquez, *Magnetic stimulation as a promoting factor for germination and*

development in *Guazuma ulmifolia* L. seeds

G07-014 - Yasmym Sarmiento Pereira, *Radiometric analysis of sediments and water from the Lagoons of East Fluminense, Rio de Janeiro.*

G07-016 - Gabriela Rodé de Assis da Silva, *Radiometric characterization of Erythrina verna Vell.: A study of natural radioactivity in plant tissues*

G8 - Luminescent materials

G08-003 - Marcelo Luis dos Santos Filho, *Development of $\text{CaSO}_4\text{:Li}$ Composites by Different Synthesis Routes for Applications in TL and OSL Dosimetry*

G08-005 - Brunno Florencia de Oliveira, *Dosimetric Potential of Cowrie Shell Biomass under High Radiation Doses using TL and OSL Techniques*

G08-017 - Caroline Pereira dos Santos, *Uncovering the intrinsic nature of the ternary metallic alloys Cu-NiPd and Cu_2PPd_2 : A DFT study*

G08-019 - André Luiz Tavares e Silva, *Unsupervised learning for pattern recognition in K-feldspar using cathodoluminescence emission and EDX composition mapping*

G08-021 - Joel Arturo Rivera-Garcia, *Estudio de las propiedades luminiscentes del aluminosilicato de estroncio dopado con Ce ($\text{SrAl}_2\text{Si}_2\text{O}_8\text{:Ce}$) sintetizado por el método de reacción en estado sólido*

G08-022 - Rodrigo Martínez Baltezar, *Point defects study on Aluminum nitride for applications in luminescence dosimetry*

12:30 LUNCH

Applications of Thermoluminescence / Luminescent Materials Virtual Room 1 Chair: Daniel Calheiro/Rodrigo Gadelha	Luminescent Materials / Dosimetry Virtual Room 2 Chair: Marco A. Labarrios/Álvaro M. L. Gómez	Medical Physics / Ionizing and non-Ionizing / Radiobiology Virtual Room 3 Chair: Jaime T. Juárez/Bruno Gallotti
13:30 Sandrith M. Pérez Ramos G01-003 - <i>Análisis cinético de la curva de brillo termoluminiscente del Óxido de Berilio (BeO) irradiado con Rayos X en un rango de baja dosis</i>	13:30 Caroline Pereira dos Santos G08-018 - <i>Unraveling the physical properties of pure and lithium thulium ion-doped calcium sulfate</i>	13:30 Erika Patricia Azorín Vega G04-005 - <i>Dosimetric and Biological Characterization of a 3D Cell Culture Model for the Evaluation of</i>
13:45 Katherine G. Delgado G01-005 - <i>Effective dose received by Dentistry students in intraoral dental radiology using TLD dosimetry</i>	13:45 Carina Oliva Torres Cortés G08-023 - <i>Repuesta Termoluminiscente ante rayos gamma de la HAp pura y dopada con tierras raras (Dy^{3+} y Eu^{3+}) a dos concentraciones 0.1 M y 0.5 M</i>	<i>Radiopharmaceutical Induced Bystander Effect</i>
14:00 Nayely D. M. Ramirez G01-008 - <i>Synthesis of Y_2O_3 nanophosphors and evaluation of their thermoluminescent response for use in radiation dosimetry</i>	14:00 Pedro R. González G08-024 - <i>Thermoluminescent properties of MgO:Sm,Tb+PTFE dosimeters</i>	13:45 Leticia González Zamora G04-010 - <i>Study of Prefractal Antennas for Multiband Radiofrequency Applications</i>
14:15 Abel Fernando Martínez Cruz G01-009 - <i>Thermally Stimulated Luminescence of Chemically Modified Magnesium Oxide with Gd^{3+} and Li^+, Obtained by a Glycine-based Combustion Process</i>	14:15 Jessica Paola C. Fraga G02-005 - <i>La termoluminiscencia del Grafito para su posible aplicación dosimétrica</i>	14:00 Maria L. Miranda Maciel G04-014 - <i>Calibration of radiochromic films EBT2, EBT3, EBT4 and RTQA2 and assessment of uncertainties for radiotherapy doses</i>
	14:30 Marcelo F. da Silva Santos G02-014 - <i>New Approaches in the Synthesis of $\text{CaSO}_4\text{:Tm,Li}$ Phosphors with Sugars as</i>	14:15 Elsa Bifano Pimenta G04-022 - <i>Evaluating Lung CT Protocols Using the Detectability Index: An Anthropomorphic Phantom-Based Evaluation of Photon-Counting and Conventional CT</i>

14:30 Michael A. M. Sánchez
G01-012 - *Analysis of TL half-life
Using a better approach to the
Exponential Integral*

14:45 Hanin Alali
G01-014 - *Assessment of Health
Impacts among Medical Radiation
Workers in a Tertiary Care
Hospital*

15:00 Berenice Candia-Rodríguez
G01-015 - *Caracterización
termoluminiscente del óxido de
aluminio (Al_2O_3 : C) irradiado
con diferentes fuentes de
radiación ionizante*

15:15 Radmila Sabitova
G01-016 - *Characterization of
additional irradiation positions in
the IVG.1M research reactor*

15:30 Mohammed Alkhorayef
G01-018 - *Development of a
Web-Based Tool for Single
Time-Point Absorbed Dosimetry
Calculation Using the H nscheid
Approximation*

15:45 E. Cruz-Zaragoza
G01-020 - *Optically -and
thermally stimulated luminescence
detection of irradiated oregano
(*Lippia graveolens*) polymineral*

16:00 Umme Muslima
G01-023 - *Thermoluminescence
Characterization of Natural Flake
Graphite Under 6 MeV Electron
Beam Exposure from Medical
LINAC for Dosimetric
Applications*

16:15 Gabriela Pontes Cardoso
G08-001 - *Cathodoluminescence
behavior of PVDF films doped
with $CuSO_4$ under high-dose
gamma irradiation*

16:30 J lia Carina Orf o Costa
G08-002 - *Characterizations of
 $SrMoO_4:Dy^{3+}$ powders
synthesized by hydrothermal
method*

16:45 Daniele da Silva Machado
G08-006 - *Dosimetric Properties
of nanoDot OSL Detectors for
Radiotherapy Quality Control: An
Initial Characterization and*

*Chelating Agents for Dosimetry
Applications*

14:45 Andr  Luiz Tavares e Silva
G01-025 - *A Data Science
Approach on the Study of Al_2O_3
Thermoluminescent Glow Curves
Under Varying Heating Rates*

15:00 Vin cius R. A. A. Martins
G02-055 - *Experimental
Evaluation of the Mechanical and
Dosimetric Properties of the Red
Perspex 4034 Dosimeter After 15
Years of Aging*

15:15 Rayline Leite da Silva
G02-084 - *Standardized Protocol
for the Analysis of Cutting
Samples and Lithological
Materials by Gamma
Spectrometry*

15:30 Nadia Rodrigues dos Santos
G02-049 - *Estimation of the
soil-to-plant transfer factor of
natural radionuclides in organic
crop from Pedra Branca State
Park*

15:45 Carolina Salinas Domj n
G02-090 - *Assessing
Water-Equivalence in
Gram-Negative Bacteria-Hydrogel
Radiation Dosimeters: A
Comparative Analysis with
Gram-Positive Bacterial Systems*

16:00 Rayline Leite da Silva
G07-017 - *Study of Drinking
Water Quality in Rio de Janeiro:
Assessment of Gamma Radiation
Levels and Physicochemical
Parameters*

16:15 Arthur Reis Martins
G06-015 - *Production of
Radioisotopes in Nuclear
Microreactors for Medical,
Industrial, and Scientific
Applications*

16:30 Jardel Lemos Thalhofer
G02-010 - *Assessment of the level
of natural radioactivity in the
region of the future Municipal
Environmental Protection Area
(APA) Ilha Verde, Muriqui,
Mangaratiba - RJ*

16:45 Leonardo da Silva Boia

14:30 Estefania Reyes Soto
G04-002 - *Automated
Classification and Manual
Segmentation of Pediatric Brain
Tumors Using Deep Learning on
MRI Data*

14:45 Giulia Rita de Souza Fa s
G04-035 - *Radiotherapy Planning
with VMAT in Halcyon 2.0:
Impacts of Isocenter Configuration
on Dosimetric Parameters and
Treatment Efficiency*

15:00 Priscila Santos Rodrigues
G04-036 - *Three-dimensional dose
mapping of gold-198
nanoparticles: A comparative
study of 3D-printed matrices using
Fricke xyleneol gel and Monte
Carlo simulations*

15:15 Hericka O.
Kenup-Hernandes
G05-004 - *Computational
Modeling for Radiological Risk
Assessment: Simulation of Plume
Dispersion in Radioactive Waste
Fires Using HotSpot*

15:30 Bruna Gomes Ara jo
G05-009 - *Gadolinium Adsorption
by Functionalized Mesoporous
Silicas (MCM- 41/SBA-15): A
Proposal for Volume and Dose
Reduction in Radioactive Waste*

15:45 Kuanysh Samarkhanov
G05-016 - *Radiological
Characterization and Dosimetric
Assessment of Liquid Radioactive
Waste from BN-350 Reactor*

16:00 Jonathan O. dos Santos
G06-017 - *Response Function
Validation of HPGe Detector
Through GADRAS and MCNP
Simulation*

16:15 Fatima Casta eda Ortiz
G07-001 - *Pulsed X-ray-Induced
Nuclear Fragmentation in a
Radioresistant Ovarian Cancer
Cell Line*

16:30 Daniele G. Mesquita
G07-006 - *Gamma Radiation
Modulates Genomic Maintenance
Pathways in T-Lymphocytes: An
Analysis of Clinical Doses*

Uncertainty Analysis

17:00 Daniel Santos Rodrigues
G08-011 - *Properties of*
CaSO₄:Eu,Mn for Luminescent
Dosimetry

G02-036 - *Construction of the*
Glioblastoma scenario in the right
optic nerve region using DIP
techniques for simulation in the
MCNP code with conversion to
lattice structure by SCAN2MCNP

17:00 Leonardo Santiago Melgaço
Silva
G02-041 - *Development of an*
optical sensor prototype for dose
readings in Fricke gel dosimeters

16:45 Walewska Gouveia Silva
G07-012 - *Proposal To Study the*
Therapeutic Potential of the
Ionizing Effect of Low-Dose
Gamma Radiation in a Murine
Model of Alzheimer's Disease

17:00 Francisco Torres Hoyos
G04-003 - *Breast carcinoma,*
Fractals, Ballistic model,
Roughness, Scaling, Cluster

13:30 to 16:00 TECHNICAL VISIT

Chair: Camila Engler

Places to visit at CDTN:

- Gamma Irradiation Laboratory - [More information, access here](#)
- Dosimeter Calibration Laboratory - [More information, access here](#)
- TRIGA IPR-R1 Reactor - [More information, access here](#)

Thursday, 09/25/2025

Invited Speaker - Lumina Auditorium

Chair: Bernardo Dantas

08:00 *Invited Speaker* – Luciana Carvalheira,

Neutrons, Knowledge, and New Generations: How Brazil's Argonauta Reactor is Shaping the Future of Radiochemistry and Its Impact on Global Dosimetry Challenges.

09:00 *Invited Speaker* – Bernardo Maranhão Dantas,

Diseminación de técnicas de monitoreo in vivo ocupacional en una Red de Clínicas de Medicina Nuclear.

09:25 *Invited Speaker* – Maria Jose Neves,

Range Radiation Exposure Indicator.

10:00 COFFEE BREAK

13:30 - 17:00 - Poster Session (Main Hall)

Note: 14:30 to 17:00 - Referees for evaluation

In-Person

G2 - Dosimetry (environmental, personal, internal, external, computational, etc.)

G02-081 - Felipe Mamed de Souza, *SMS Notification System for Potential Radiological Risk Scenarios*

G02-085 - Alcilene Cristina da Silva, *Study and Characterization of the EPR Response of New Materials for High Dose Dosimetry*

G02-088 - Ana Letícia Almeida Dantas, *The IRD In Vivo Monitoring Laboratory - LABMIV: Instrumental Capabilities applied in routine and emergency situations*

G02-091 - Maryanna Regina de Souza Roberto, *OSL Tandem Dosimeter for ICRU 95 Operational Quantities*

G02-092 - Thalys Matheus Gama de Oliveira, *Photons Response of an Albedo Monitor*

G3 - Ionizing and non-ionizing radiation

G03-001 - Paula Regina Rodriguez Coelho, *"Analysis of spatial resolution in two different phantoms as a quality parameter in MRI pediatric pathology"*

G03-002 - José Marques Lopes, *Adjustment Factor for Gamma Spectra Due to the Influence of Sample Density on the Shielding Background Radiation*

G03-003 - Davyd Soares Dunda, *Application of Steam Methodology in Science Teaching: Prototyping a Cloud Chamber*

G03-004 - José Marques Lopes, *Comparison of environmental sampling results using portable and benchtop scintillation detectors*

G03-006 - Adriana de Souza Medeiros Batista, *Effect of repolymerization and gamma irradiation on the morphology of styrene-divinylbenzene composites with magnetite*

G03-007 - Paulo Cezar Rocha Silveira, *Gamma Irradiation Effects on the Mechanical and Ballistic Performance of 3D-Printed Polylactic Acid*

G03-008 - Luciana Carvalheira, *Gold nanoparticles with potential for use in multimodal combat of head and neck cancer*

G4 - Medical Physics

G04-006 - Otávio Gabriel Moreira Silva, *SNR and CTDIvol study as a function of the Noise Index in Computed Tomography exams*

G04-016 - Rute Cristina de Oliveira Soares, *Chest Computed Tomography in Oncology: Clinical Contributions and Dosimetric Assessment*

G04-017 - Gustavo Freire Pereira da Silva, *Comparative Validation of the DEEDZ Methodology and a Photon-Interaction-Based Model Implemented in TEMPy for Estimating Effective Atomic Number and Electron Density*

G04-018 - Rute Cristina de Oliveira Soares, *Contributions of Computed Tomography to Breast Cancer Staging and Follow-Up: Justification of Practice and Dosimetric Analysis*

G04-021 - Iasmin Vieira Nishibayaski, *Dosimetric evaluation of EBT3 and EBT4 in superficial HDR brachytherapy for keloids*

G04-023 - Camila Engler, *Evaluation of Breast Positioning Quality and Its Association with Technical Parameters in Screening Mammography*

G04-030 - Raissa Alexia Camargo Guassu, *Impact of Body Mass Index on Radiation Doses in Interventional Radiology*

G04-031 - Douglas Philipe Martinho Dos Santos, *Investigation of Diagnostic Reference Levels (DRL) in medical radiography: dose-area product (DAP)*

G04-033 - Roger Ferreira da Silva, *Mammography quality assurance based on phantom image with deep learning*

G04-034 - Kellen Adriana Curci Daros, *Radiation dose from pediatric coronary artery calcium scoring CT scan to evaluate chronic kidney disease-mineral and bone disorder: Observational study*

G04-038 - Arícia Ravane Pereira da Cruz, *Validation of an alanine-based postal dosimetry system for quality control in radiotherapy*

G6 - Radiation Sources

G06-013 - Kathleen Martins, *Neutronic Analysis of an Accelerator-Driven Subcritical Reactor Utilizing Thorium Dioxide and Reprocessed Nuclear Fuels*

G7 - Radiobiology

G07-003 - Christiana da Silva Leite, *Comparative Study of Organ Mass in Healthy and Tumor-Bearing Mice for Preclinical Dosimetry Purposes*

G07-005 - Camila Ramos Silva, *Development and validation of a body-shielding device for radiotherapy of breast cancer in mice using a panoramic gamma source*

G07-008 - Luciana Penedo de Melo Teixeira, *Impact of Physical Exercise on Irisin Expression in the Brain of Irradiated Mice: Effects on Behavior and Memory*

G07-009 - Olanía Herrera González, *Involvement of SMARCB1 subunit of the mSWI/SNF and GLL-1 epigenetic complex and their radiolabeling as a strategy for ionizing therapy in lung cancer*

G07-011 - Camila de Almeida Salvego, *Nitric Oxide-Donor Nanoparticles Potentiate Radiotherapy of Triple-Negative Breast Cancer*

G07-018 - Bruno Gallotti, *Growth Inhibition of *S. cerevisiae* BY4741 Exposed to Gold Nanoparticles and Low-Energy Radiation*

G07-019 - Maurício Moacir da Silva Borges, *The Role of Caffeine in Reducing TNF- α and Improving Cognitive in Irradiated Mice in the Brain*

G07-020 - Luis Martín García Ortiz, *Cellular response in human leukocytes (in vitro), due to exposure to UVB radiation using apoptotic biomarkers*

Invited Speaker - Lumina Auditorium

Chair: Telma C. F. Fonseca

10:30 Invited Speaker – Ademar Lugo, *Use of ionizing radiation for the production of nanoparticles and advanced curatives.*

11:30 Invited Speaker – Richard Hugtenburg, *New Technologies as a Route to Particle Therapy: Laser-Hybrid Acceleration for Radiobiology Applications – The LhARA Project.*

12:30 LUNCH

13:30 *Invited Speaker* – Hélio Yoriyaz,
Proton Beam Therapy: Main Characteristics and Challenges.

Dosimetry
Spectrum Auditorium
Chair: Peterson Squair

Medical Physics/Dosimetry
Lattice Auditorium
Chair: Lucas Paixão

13:30 Divanizia do Nascimento Souza
 G02-042 - *Dose Distribution in an Anthropomorphic Cervix Phantom Using Two Computer Modeling Approaches*

13:45 Francisco Harley Dantas Hauradou Xavier
 G02-057 - *Generator of Health Optimized Simulation Templates – GHOST a 3D Slicer plugin to voxelized medical images for MCNP*

14:00 Lucas Dias Rodrigues
 G02-058 - *Impact of Granulometry on the Natural Radioactivity of Granite: Health Implications and Applications in Building Materials*

14:15 Francisco H. D. Hauradou Xavier
 G02-094 - *Optimization of dose rate map acquisition time for internal dosimetry in nuclear medicine using 3D U-Net convolutional neural networks*

14:30 José Elizeu de Souza Junior
 G02-062 - *Influence of Photon Emission Spectra on Air Kerma Rate Constants for Radionuclides: A Monte Carlo EGSnrc Study*

14:45 Raphael Francisco Gomes dos Santos
 G02-063 - *Influence of Radiation Shielding on Effective Dose in Prostate Cancer Radiotherapy with a Varian Clinac 15 MV*

13:30 Ana Beatriz Caldeira de Andrade
 G04-025 - *Evaluation of spatial resolution in PET equipment with a Jaszczak-analog phantom built by additive manufacturing*

13:45 Crislane de Jesus Cesário
 G04-028 - *Experimental Evaluation of 3D-Printed Materials (ABS and PETG) for Mammographic Phantom Applications Using $\text{CaSO}_4:\text{Dy}$ Dosimetry*

14:00 Nathalia Luzia Aidar Alves
 G04-032 - *Machine Learning in PSQA: Performance of a Random Forest Model on Radiotherapy Metrics*

14:15 André Luiz Espindola Fidelis
 G04-037 - *Validation of a 3D-Printed Anthropomorphic Eye Phantom for Quality Assurance in External Beam Radiotherapy Using TLDs and Radiochromic Film*

14:30 Leonardo Gualberto Silva Macedo
 G02-095 - *Mass Analysis of Computational Mouse Models Developed at CDTN: A Comparative Study of Male and Female Versions*

14:45 Beatriz Diniz de Oliveira Guedes
 G02-082 - *Spectrophotometric Analysis of a 3D-Printed Photopolymerizable Resin Exposed to High-Dose Gamma Radiation*

15:00 Hirys Sales
 G02-053 - *Evaluation of Polymeric Materials as Muscle Tissue Equivalents for Physical Phantoms Used in the Calibration of In Vivo Monitoring Systems*

14:30 *Invited Speaker* – Georgia Joana,
Win Brasil - Gestão Diversidade & Inclusão.

15:30 COFFEE BREAK

16:00 - Mentorship Helen Khoury WIN Brasil (Spectrum Auditorium)

Empowering the Future of Nuclear Energy: A Mentorship Program for the Development of Female Leadership
Moderator: Ketrim souza,
Panelists: Georgia Joana and Winners (Winbrasil)

Roundtable Session (Lumina Auditorium)

Chair: Denison Santos

16:00 Leadership in Scientific and Technological Research

Moderator: Luciana Carvalheira,

Panelists: Adriana Marques and Anna Letícia Barbosa de Sousa

**Radiation Protection
 Lumina Auditorium
 Chair: Teógenes da Silva**

**Radiation Sources / Radiation Protection
 Lattice Auditorium
 Chair: Jhonny Benavente**

17:00 Raphael Francisco Gomes dos Santos

G05-005 - *Determination of Radiation Protection*

*Procedures for an Offshore Plant Inspection in Brazil
 Using Radiotracer Techniques*

17:00 Gabriel do Nascimento Freitas

G06-014 - *Nuclear simulation workflow to correct
 gamma ray well logs for radioactive contaminated
 drilling fluids*

17:15 Mirella Maturo da Cruz

G05-021 - *Transfer Factors of Natural Radionuclides
 in Organic Foods from Rio de Janeiro, Brazil*

17:15 José Alípio dos Santos Filho

G06-016 - *Proposal for the construction of a neutron
 generator for application in research and industry*

17:30 Elydio J. D. Soares

G05-010 - *Investigation of radon exhalation from
 mortars made from niobium tailings and the
 mitigating role of coating materials*

17:30 Welen Nunes de Lima

G05-014 - *Proposal for the use of thorium as fuel in
 TRIGA® research reactors*

Friday, 09/26/2025

Invited Speaker - Lumina Auditorium

Chair: Richard Hugtenburg

08:00 *Invited Speaker* – Rolf Behrens,
From dosimeter development to routine use – Standards and Uncertainties.

08:40 *Invited Speaker* – Susana Lalic,
Plant Stem Cells as an Ethical Alternative for Biodosimetry.

09:20 *Invited Speaker* – Thiago Viana Miranda Lima,
Current and Future Trends in Nuclear Medicine Dosimetry.

10:00 COFFEE BREAK

**Radiological Sources
Spectrum Auditorium
Chair: Carmen C. Bueno**

**Dosimetry
Lumina Auditorium
Chair: Peterson Squair**

10:30 Matheus R. do Nascimento
G03-009 - *Speknife: open-source software for X-ray spectra correction in solid state dosimetry*

10:30 Jhonny A. Benavente Castillo
G02-007 - *Simulación del blindaje del búnker de un acelerador ciclotrón utilizando el código MCNP*

10:45 Eduardo B. de Paula
G05-011 - *Metrological Enhancement of Radiopharmaceuticals Half-Lives Determination via γ -ray Spectrometry and Algorithmic Data Processing*

10:45 Natasha Briggs e Silva
G02-011 - *Preliminary Evaluation of a Positioning Phantom for Dosimetric Audit in Ir-192 HDR Brachytherapy: Results from an IAEA Collaborative Project*

11:00 Igor Andrade Machado
G02-083 - *SpecUnPy code: updates and what is coming*

Invited Speaker - Lumina Auditorium

Chair: Bruno M. Mendes

11:30 *Invited Speaker* – Tarcísio P. R. de Campos and Alberto Mizrahy Campos,
Challenges in Radioisotope Decontamination and Production through Neutron Spectral Selection.

12:30 LUNCH

**Dosimetry
Spectrum Auditorium
Chair: Maria da Penha A. Potiens**

**Dosimetry
Lattice Auditorium
Chair: Bruno M. Mendes**

**Applications of Thermoluminescence
Lumina Auditorium
Chair: Teógenes da Silva**

13:30 Isabela N. S. Ferreira
G02-064 - *Mapping of natural radionuclides U, Th and K in the city of Belo Horizonte/Minas Gerais*

13:30 Celso G. de Paulo
G02-069 - *Monte Carlo simulation of a HPGe detector*

13:30 Tarcísio P. R. Campos
G01-046 - *The reduction of the radiological impact of nuclear fuel waste*

13:45 Claudio A. Federico
G02-065 - *Measurements of Ionizing Radiation Dose Onboard an Aircraft from Southeastern*

13:45 Ana G. Leão Ferro
G02-075 - *Occupational exposure assessment in ROLL procedures with Tc-99m: A Monte Carlo simulation approach*

14:00 Júlia B. Severo
G01-039 - *Environmental Gamma Radiation and Radon Exposure in Areas Surrounding Urban Quarries in Belo Horizonte,*

Brazil to the Antarctic Continent **14:00** Rafael S. Oliveira *Brazil*
14:00 Paulo J. Iack Silva G02-076 - *On the adaptation of a*
G02-068 - *Monte Carlo N-Particle* *commercial Bonner Sphere*
(MCNP) Simulations for *Spectrometer to be used with*
Scintillation Emission in *thermoluminescent detectors*
Quantum Dot-Doped Systems

Invited Speaker - Lumina Auditorium

Chair: Telma C. F. Fonseca

14:30 *Invited Speaker* – Yvone Maria Mascarenhas,
Personal Dosimetry: The Importance of Regulation, Technology, and Culture. Evaluation of Dosimetry
Services in Latin America and the Caribbean.

15:30 COFFEE BREAK

Invited Speaker - Lumina Auditorium

Chair: Telma C. F. Fonseca

16:00 *Invited Speaker* – Carmen Bueno,
Radiation processing with electron beams: On the feasibility of using electronic dosimeters for real-time dosi-
metry.

17:00 CLOSING SESSION

- Final Session: Summary and Perspectives: Telma C. F. Fonseca & Bruno M. Mendes
- Honorary Award in Memory of Helen Jamil Khoury: Carmen Bueno
- Helen Khoury Award for Best Works: Andrea Mantuano & Ester Andrade